

A smart mushroom incubation system for enhanced yield and sustainability in
small scale farming

RP 25-26J-264

Project Proposal Report

IT22062888

G. A.I.W. Wijesinghe

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology
Faculty of Computing
Sri Lanka Institute of Information Technology

August 2025

A smart mushroom incubation system for enhanced yield and sustainability in
small scale farming

RP 25-26J-264

Project Proposal Report

IT22062888

G. A.I.W. Wijesinghe

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology
Faculty of Computing
Sri Lanka Institute of Information Technology

August 2025

Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.



Signature

29/08/2025

Date

Signature of the Supervisor

.....
Date

Signature of the Co-Supervisor

.....
Date

Abstract

The purpose of this research is to improve the reliability and sustainability of oyster mushroom farming in Sri Lanka through the introduction of a Smart Mushroom Incubation System [1][2]. Mushroom cultivation is widely practiced across the country, particularly by small-scale farmers, because of its relatively low initial investment and high nutritional value [3]. However, the industry continues to face several persistent challenges that reduce both yield and quality. Manual monitoring of environmental conditions requires constant attention and is often inaccurate, leading to unfavorable growth environments [4][5]. In addition, inconsistent yields from one cycle to another, combined with the risk of pest infestations, result in financial losses and discourage many farmers from expanding their operations [6][7]. These problems highlight the need for a modern, technology-driven solution that reduces labor demands while improving efficiency and crop health [8].

To address these issues, this study proposes the design and development of an IoT-based Smart Incubation System tailored for oyster mushroom cultivation [1][9]. The system integrates a network of sensors to continuously monitor critical environmental factors such as temperature, humidity, and carbon dioxide concentration [2][5]. Data is collected in real-time and made accessible through a mobile-friendly dashboard, which provides farmers with actionable insights and instant alerts when conditions move beyond the ideal range [4][9]. By enabling early identification of unfavorable changes and potential pest risks, the system allows farmers to respond quickly and maintain stable growing conditions without relying on constant manual observation [6][8].

The findings suggest several important benefits. Automated monitoring reduces dependence on manual labor, saving farmers time and effort while lowering the chances of human error [1][4]. More consistent environmental control contributes to steady yields and improved crop quality, helping farmers secure better prices in the market [2][6]. Additionally, by alerting farmers to potential risks at an early stage, the system minimizes crop losses due to pests and reduces the unnecessary use of chemical pesticides, supporting healthier and more sustainable farming practices [7][10].

In summary, the Smart Mushroom Incubation System represents a practical, low-cost solution for improving the efficiency and profitability of mushroom farming in Sri Lanka [1][2][8]. By combining traditional farming knowledge with modern IoT technology, this research demonstrates how small-scale farmers can enhance productivity, reduce risks, and move toward more sustainable agricultural practices [3][6][9]. In the long term, the adoption of such systems has the potential to transform mushroom cultivation into a more reliable, technology-driven industry that strengthens rural livelihoods and contributes to food security in the region [5][10].

Table of Contents

<i>Declaration</i>	<i>1</i>
<i>Abstract</i>	<i>4</i>
<i>Introduction</i>	<i>6</i>
<i>Background & Literature survey</i>	<i>6</i>
<i>Research Gap</i>	<i>9</i>
<i>Research Problem</i>	<i>11</i>
<i>Objectives</i>	<i>12</i>
Main Objectives	<i>12</i>
Specific Objectives	<i>12</i>
<i>Methodology</i>	<i>13</i>
<i>Project requirements</i>	<i>14</i>
1. Functional requirements	<i>14</i>
2. Non-Functional Requirements.....	<i>14</i>
3. User requirements	<i>15</i>
4. System requirements.....	<i>15</i>
<i>References</i>	<i>18</i>
<i>Gantt Chart</i>	<i>Error! Bookmark not defined.</i>

Introduction

Mushroom cultivation is an emerging agricultural sector in Sri Lanka, providing farmers with a profitable and sustainable source of income [1][2]. Oyster mushrooms, in particular, are widely cultivated due to their nutritional value, short growth cycle, and high demand in local and international markets [3]. However, mushroom farms are highly sensitive environments that require strict control of temperature, humidity, and hygiene [4]. Among the major threats to productivity are pest infestations, which can rapidly damage crops, lower yields, and reduce quality [5]. Effective pest management is therefore critical to sustaining healthy cultivation and ensuring consistent harvests [6].

Traditionally, farmers rely on manual inspection to detect pest activity inside mushroom farms [2][7]. This process is not only time-consuming but also unreliable, as many pests are active at night or in hard-to-reach areas [5]. More importantly, human entry into the mushroom house increases the risk of contamination, since mushrooms are highly vulnerable to bacteria, fungi, and other pathogens carried by people [4][8]. Conventional pest control methods, such as chemical spraying, are often applied reactively after an infestation is detected, leading to delays and inconsistent results [6][9]. These limitations highlight the need for an automated, real-time solution that minimizes human involvement while protecting the sensitive cultivation environment [1][8].

Recent advances in smart farming and computer vision have paved the way for innovative pest management systems [7][10]. State-of-the-art approaches in agriculture often use fixed cameras, IoT sensors, or manual image capture combined with machine learning to identify pests in fields or greenhouses [3][6]. For example, some studies have employed static camera traps or sensor networks to monitor insect behavior, while others have used drones to capture aerial images of large-scale farms [9]. Although these methods show promise, they are less effective in mushroom farming environments, where space is limited, pests move quickly, and blind spots can easily occur [5][8]. Moreover, most existing systems stop at detection and do not integrate automated response mechanisms, leaving farmers to take manual action [7][10].

The proposed Pest Management Module introduces a novel approach by combining a moving camera system with automated pest control mechanisms [1][2]. Unlike static systems, the moving camera can cover the entire mushroom farm, reducing blind spots and ensuring continuous surveillance [5][7]. The images are processed using a Convolutional Neural Network (CNN) model to detect pests in real time with high accuracy [3][9]. Once pest activity is identified, the system automatically activates a neem-based spraying mechanism to control the infestation immediately, while also sending alerts and detailed insights to farmers through a Flutter mobile application [4][6]. The backend logic is powered by Spring Boot, image detection is handled by Python, and Firebase is used for instant notifications and real-time updates [8][10].

What makes this approach different from previous solutions is the integration of detection, automated response, and farmer notification into a single, seamless system [2][7]. While existing solutions primarily focus on monitoring or early warning, this system not only detects pests but also takes direct action to manage them, reducing delays and minimizing farmer entry into the sensitive farm environment [1][9]. This reduces contamination risks, ensures timely pest control, and provides valuable insights into pest activity patterns, thereby enhancing both yield quality and sustainability in mushroom farming [5][10].

Background & Literature survey

Mushroom farming, especially oyster mushroom cultivation, has become an important livelihood activity for many small-scale farmers in Sri Lanka and other developing countries, mainly because it requires relatively low initial investment, shorter growth cycles, and provides a good return compared to other agricultural practices. Beyond its economic benefits, mushrooms also contribute to food security by offering a rich source of protein, vitamins, and minerals at an affordable cost. However, despite these advantages, mushroom farming faces a series of persistent challenges, among which pest infestations remain one of the most critical. Pests can cause severe crop losses within a short time, reducing both yield and quality, and in some cases destroying entire harvests. The difficulty is further compounded by the fact that mushroom farms are highly sensitive environments where temperature, humidity, and cleanliness must be carefully maintained. Human entry into the farm for regular inspections, though necessary in traditional systems, carries the risk of introducing harmful germs and pathogens, which in turn increases the likelihood of contamination and crop failure. This dependence on manual checks not only increases labor requirements but also leaves farmers vulnerable to delayed responses, since pests are often active at night or in hidden areas where they may go undetected. Current practices such as scheduled spraying or reactive pest control measures are inconsistent, slow, and sometimes excessive, leading to unnecessary costs, reduced efficiency, and potential harm to the environment.

Over the past decade, advancements in smart farming and precision agriculture have attempted to address similar problems in open fields and greenhouses through the use of sensor networks, IoT devices, and computer vision. Researchers have experimented with static camera systems, insect traps equipped with sensors, and even drones to detect pest activity, supported by machine learning models like Convolutional Neural Networks (CNNs) to identify insects from captured images. While these approaches demonstrate progress, they are not fully suitable for mushroom farms where space is restricted, conditions are delicate, and pest activity is often sporadic and localized. Moreover, most of these systems remain limited to the detection stage and still depend on farmers for manual intervention, creating

delays between identification and actual pest control. This gap highlights the need for a more integrated solution that not only monitors but also responds to pest threats automatically.

The proposed Pest Management Module introduces a novel approach that combines continuous monitoring, real-time detection, and automated response into one unified system. Unlike traditional fixed monitoring devices, the system uses a moving camera to cover the entire mushroom house, thereby reducing blind spots and ensuring that pest activity is captured wherever it occurs. Images collected from the camera are analyzed using a CNN model to identify pests with high accuracy, while the backend logic managed by Spring Boot processes detections and coordinates automated responses. When pests are detected, a neem-based spraying mechanism is immediately activated to control the outbreak, and farmers are notified through instant alerts delivered via a Flutter mobile application integrated with Firebase. This end-to-end workflow transforms pest management from a reactive, manual process into a proactive and automated solution that minimizes farmer entry, reduces contamination risks, and ensures timely interventions.

The development of such a system contributes new knowledge to the field of smart farming by demonstrating the potential of combining moving camera surveillance with automated control measures, a feature rarely integrated in previous studies. From a societal perspective, the Pest Management Module is highly significant as it directly addresses issues that threaten the income and security of small-scale farmers. By reducing labor demands, preventing large-scale crop losses, and minimizing unnecessary chemical use, the system supports both economic stability and sustainable farming practices. Furthermore, it empowers farmers with data-driven insights into pest activity patterns, helping them make informed decisions about farm management. In the broader context, solutions like this contribute to strengthening food security, promoting environmentally friendly agriculture, and modernizing traditional farming practices in Sri Lanka, setting an example that can be adapted to other regions facing similar challenges.

Research Gap

Research A – Real-Time Agricultural Monitoring with Agrobot:

This study presents Agrobot, a robotic system for real-time detection of plant diseases and pests, focusing on tomato crops in Sudan. Using a Raspberry Pi controller and a Pi camera, the robot scans crops while servo motors allow comprehensive movement of the camera. The system uses YOLO object detection models to identify major diseases like late blight and Septoria leaf spot, as well as pests such as the fruit borer. A comparison between YOLOv4 and Tiny YOLOv4 was conducted to evaluate accuracy and efficiency. Results indicate that Agrobot can effectively detect tomato diseases and pests, enhancing crop management and improving yield.

Research B – Insect Pest Detection and Identification Method Based on Deep Learning:

This research proposes a two-stage deep learning method for detecting and identifying small insect pests in agricultural fields. Using YOLOv3 as a region proposal network for initial detection and Xception CNN for re-identification, the approach addresses the challenge of detecting small objects that are visually similar. The study also introduces a data augmentation method to train the models effectively, overcoming the scarcity of large-scale pest datasets. The proposed system enables automatic monitoring of insect pests, supporting pest forecasting and management for improved agricultural productivity.

Research C – Image Detection and Recognition of Different Species of Animals Using Deep Learning:

This paper explores the use of YOLOv5 for real-time detection and recognition of multiple animal species. The system can process images, live camera feeds, and recorded videos to detect animals like horses, cows, elephants, birds, and more. YOLOv5 enhances object detection efficiency through improvements such as multiclass logistic regression, the Focus module, and CSPNet modules for feature fusion. The model automatically learns bounding box anchors and uses the GIoU loss function to improve detection accuracy. This approach provides fast, accurate animal monitoring, useful for wildlife management, livestock tracking, and environmental monitoring.

Application Reference	Identification of CCI (Crop Condition Indicators)	Progression Level Detection of CCI	Real-time Monitoring	Automatic Response	Mobile Integration
Research A	Damaged leaf area ✓ / Number of caterpillars X / Mobile-based identification X	X	X	X	X
Research B	X	X	X	X	X
Research C	X	X	X	X	X
Proposed System	✓	✓	✓	✓	✓

Research Problem

In mushroom farming, one of the most pressing challenges is the lack of a reliable and non-intrusive method to detect pest activity [3][5]. Farmers are often forced to physically enter farms to check mushroom bags, which not only consumes valuable time but also risks introducing harmful germs and disrupting the delicate growing environment [4]. Manual inspections are inconsistent, depending heavily on human judgment, and often allow pests to spread before being noticed [2]. The shortage of advanced tools designed specifically for mushroom farms further amplifies this issue, leaving growers with limited and outdated practices for pest management [6]. Misidentifying or overlooking pests during manual checks can quickly escalate into infestations, causing significant yield losses and increasing the need for excessive chemical use [7].

This research aims to address these issues by developing a moving camera platform that patrols between mushroom rows, capturing images from multiple angles and applying real-time image processing to detect pests instantly [8]. Farmers would receive immediate alerts through a mobile or web application, complete with pest location details, enabling them to respond without entering the farm [1]. To further strengthen control, the system integrates an automatic neem spraying mechanism that activates only in the affected areas, reducing chemical overuse and protecting the natural balance of the farm [9]. Over time, the system will analyze pest patterns and hotspot locations, offering farmers insights that can improve farm layout and preventive strategies [10].

The impact of developing this solution is significant: it minimizes the risks of contamination, reduces farmer workload, ensures timely responses, and optimizes chemical usage [2]. Beyond immediate pest control, the system provides valuable knowledge about how and where pests typically occur, supporting smarter long-term management [4]. Ultimately, this component seeks to transform pest monitoring from a reactive, manual process into a proactive, automated, and eco-friendly solution—helping farmers safeguard both their crops and their livelihoods [7][8].

Objectives

The overall purpose of this study is to design and implement a **Smart Pest Detection and Automated Response System** that overcomes the limitations of traditional manual pest monitoring in mushroom farms. The research aims to reduce contamination risks, improve detection accuracy, save farmers' time, and ensure eco-friendly pest control. By combining a mobile camera platform, real-time computer vision, automated neem spraying, and data-driven insights, the system seeks to transform pest management from a manual, reactive task into a proactive, automated process.

Main Objectives

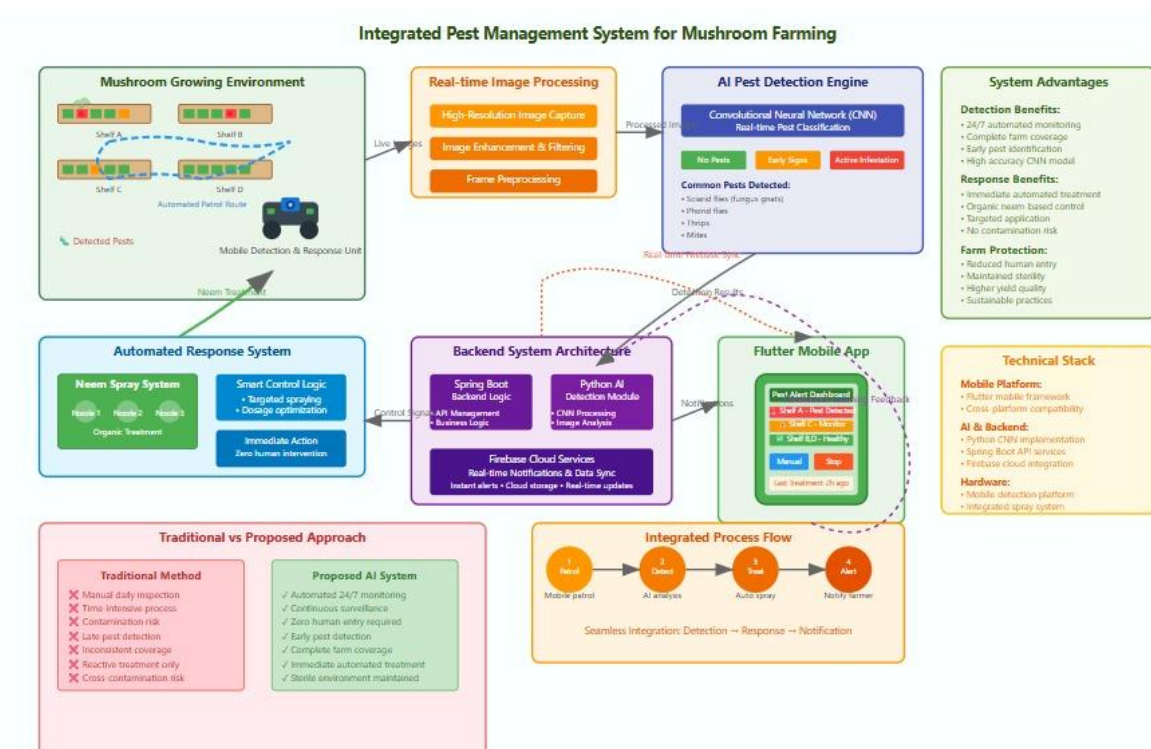
To develop an automated pest detection and control system for mushroom farms using a mobile camera platform that patrols along mushroom racks, captures images of bags row by row, applies real-time computer vision for pest identification, alerts farmers through a mobile/web app, and activates targeted neem spraying in affected areas minimizing human entry, reducing contamination risks, and optimizing chemical usage.

Specific Objectives

- **To design a mobile camera platform** that moves along the aisles between mushroom racks, capturing clear front and side views of mushroom bags during patrols, ensuring maximum possible coverage without disturbing the farm environment.
- **To implement real-time computer vision algorithms** capable of detecting pest activity or unusual movements on mushroom bags, even under challenging farm conditions such as dim lighting, high humidity, and repetitive bag patterns.
- **To develop a mobile/web application** that delivers instant pest alerts with details such as the pest's location (rack number/row), images, and detection confidence, enabling farmers to take quick action without physically entering the farm.
- **To integrate an automated neem spraying mechanism** that activates only in zones where pests are detected, ensuring targeted treatment while minimizing unnecessary chemical use and protecting the mushrooms' quality.
- **To establish a pest activity logging and analysis system** that records the frequency, timing, and location of pest appearances. This data will help identify recurring hotspots, seasonal trends, or farm layout inefficiencies that contribute to pest infestations.
- **To evaluate the system in real farm settings** by measuring detection accuracy, spraying precision, response time, and user-friendliness, ensuring that the solution is practical, reliable, and easily adoptable for mushroom farmers.

Methodology

The methodology for this study focuses on designing and implementing a Smart Pest Detection and Automated Response System for mushroom farms. The system development begins with the design of a mobile camera platform capable of patrolling between mushroom rows, capturing continuous images from multiple angles to ensure complete coverage and minimize blind spots [5][6]. Real-time image processing and computer vision algorithms are then applied to these images to detect pest activity or unusual movements on mushroom bags [1][2][4]. When a pest is detected, the system immediately sends an alert to farmers via a mobile or web application, providing precise location details and visual evidence [3][8]. Simultaneously, the detection system triggers an automated neem spraying mechanism that activates only in affected areas, reducing chemical overuse and contamination risks [7][9]. Collected data on pest frequency, location, and patterns is stored and analyzed to generate insights for farm layout optimization and proactive pest management [8][10]. The methodology emphasizes iterative testing and calibration of the platform and algorithms to ensure high detection accuracy under varying lighting and environmental conditions, making the system reliable, efficient, and practical for real-world mushroom farm operations [1][5].



Project requirements

The project will begin with a detailed planning and requirements gathering phase. This stage will involve consultations with key stakeholders, including mushroom farmers, agricultural experts, and technology specialists, to identify the specific challenges in pest management and the expected outcomes of the proposed system. Based on these discussions, the requirements of the Smart Pest Management Module have been categorized into functional requirements, non-functional requirements, system requirements, and user requirements.

1. Functional requirements

- **Real-Time Pest Detection** – The system must continuously monitor live video feeds from the moving camera to detect pest activity within the mushroom farm.
- **Automated Pest Control** – Upon detection, the system should automatically trigger neem spraying to control pests without requiring farmer intervention.
- **Image Analysis with CNN** – Captured images must be analyzed using a CNN model to accurately identify pests and distinguish them from harmless movements.
- **Alert and Notification System** – The system must send instant alerts to farmers via the Flutter mobile application, including images and details of detected pest activity.
- **Farm Insights Dashboard** – The mobile app should display real-time data, pest activity history, and environmental insights to help farmers make informed decisions.
- **Mobile App Integration** – The app must allow farmers to receive notifications, view pest detection records, monitor automated responses, and manage farm profiles.
- **Data Storage and Management** – All pest detection data, images, and spraying records must be stored securely in Firebase for future reference and analysis.

2. Non-Functional Requirements

- **Performance** – The system must process live video streams and detect pest activity with minimal delay to ensure timely responses.
- **Reliability** – The module should operate consistently under continuous monitoring, with minimal downtime, to avoid missing pest activity.
- **Scalability** – The system must be able to handle multiple cameras or larger mushroom farms without performance degradation.
- **Security** – All captured data, images, and farmer details must be stored securely in Firebase, with controlled access to protect sensitive information.
- **Usability** – The mobile application should provide a simple, intuitive interface so that farmers with limited technical knowledge can easily operate the system.
- **Maintainability** – The system should be modular and well-documented, allowing future upgrades or modifications without major redesign.

- **Portability** – The solution must be compatible across different mobile devices (Android/iOS) and adaptable to various mushroom farm environments.
- **Accuracy** – The CNN-based detection system should maintain high accuracy in identifying pests while minimizing false positives and negatives.
- **Efficiency** – The automated spraying mechanism should optimize neem usage, avoid waste while ensuring effective pest control.
- **Availability** – The system must always be accessible to farmers, with real-time updates and alerts delivered reliably through the mobile app.

3. User requirements

- Farmers must be able to register their farm details by uploading basic information (farm size, type of mushroom cultivated, and location).
- Users should receive timely notifications and alerts when pests are detected, including captured images or video evidence for verification.
- Farmers must be able to view pest detection history, including past alerts, spraying activities, and effectiveness reports.
- Users should have access to a mobile app where they can monitor live farm footage, check real-time pest status, and manage automated spraying.
- The system should allow users to control or override the spraying mechanism manually through the mobile application when required.
- Farmers must be able to securely store and access their farm-related data in the system for future analysis and planning.
- Users should have access to simple guides or system-generated recommendations on pest prevention and mushroom farm safety.

4. System requirements

Hardware Requirements

- **Mobile Camera Unit**
 - Capable of capturing high-resolution images of mushroom bags from multiple angles.
 - Portable and lightweight to move along cultivation rows automatically or manually.
 - Durable to withstand farm conditions, including humidity, temperature variations, and possible dust exposure.
- **Actuation Unit (Automated Sprayer)**
 - Mechanism for precise neem spraying in affected zones.
 - Controlled via software to activate only where pests are detected, reducing chemical overuse.

- Safe and reliable for repeated operations near mushrooms.
- **Processing Unit**
 - Embedded processor or mobile device capable of running real-time image processing and computer vision algorithms.
 - Sufficient memory and storage to handle continuous image capture and analysis without lag.
- **Storage**
 - Local storage to temporarily save captured images, detection logs, and spray actions.
 - Optional cloud storage for long-term data retention, historical analysis, and system updates.
- **Display Devices**
 - Mobile phone, tablet, or desktop computer to access real-time alerts and farm dashboard.
 - Interface displays pest location, images, and spray status for actionable insights.
- **Power Supply**
 - Rechargeable battery or plug-in solution sufficient to power the camera, processor, and sprayer for full operation periods.
 - Backup power option to ensure continuous monitoring during unexpected outages.

Software Requirements

- **Image Processing and Computer Vision**
 - Lightweight models capable of detecting pests in real-time under variable lighting and angles.
 - Algorithms for movement detection and anomaly recognition on mushroom bags.
- **Mobile/Web Application**
 - Dashboard to send instant alerts to farmers with pest location, images, and confidence scores.
 - User-friendly interface for monitoring and controlling the sprayer.
- **Data Management**

- Logging module for storing pest detection events, spray actions, and timestamps.
- Optional analytics for identifying pest frequency patterns and hotspots.
- **Connectivity**
 - Wi-Fi or local network connectivity for sending alerts and synchronizing data between the camera platform and dashboard.
 - Optional offline functionality with local storage in case of network unavailability.

References

[1] **Albanese, Nardello & Brunelli (2021):** *Automated Pest Detection with DNN on the Edge for Precision Agriculture*

Demonstrates using a low-power embedded system with deep neural networks to detect pests autonomously in fruit orchards—important foundation for low-latency, real-time inference on mobile platforms.

[2] **Ung, Ung & Nguyen (2021):** *An Efficient Insect Pest Classification Using Multiple Convolutional Neural Network Based Models*

Presents advanced CNN architectures for accurate pest classification on benchmark datasets, showing high precision and relevance to your detection algorithms.

[3] **Jafar et al. (2024) / App-based Tools (Review):** *Smartphone-Based Citizen Science Tool for Plant Disease and Insect Pest Detection Using Artificial Intelligence*

Reviews the development and effectiveness of mobile AI apps in detecting plant pests, emphasizing the importance of mobile interfaces, reliability, and user trust.

[4] **Ultralytics (Pest Detection for Greenhouses):** *How smart greenhouses use computer vision for early pest detection*

Highlights real-time CV pest detection using pre-trained YOLOv8 models—applicable to your system's model selection strategy.

[5] **Sensors (2022):** *Deep-Learning-Based Wireless Visual Sensor System for Shiitake Mushroom Sorting*

Describes a ViT-based visual sensor system for sorting mushrooms with almost perfect accuracy—relevant for imaging and inference methods.

[6] **Farming Robotics Lab (UF):** *Automated Mushroom Detection and Maturity Assessment System*

Uses depth imaging and deep learning (YOLOv8 + SVM) for detecting mushrooms' maturity. This shows potential pathways for capturing multi-angle visual data.

[7] **Brandoli et al. (2020):** *DropLeaf: a precision farming smartphone application for measuring pesticide spraying methods*

A mobile computer vision tool to measure pesticide coverage—useful reference when calibrating and assessing your neem-sprayer accuracy.

[8] **Applied Sciences (2022):** *Internet of Things Meets Computer Vision to Make an Intelligent Pest Monitoring Network*

Combines edge CV nodes and cloud analysis for daily pest monitoring, illustrating architectures you might mirror or adapt.

[9] **Times of India (2025):** IIT Kharagpur's ground-based robot that detects plant disease and selects pesticides for spraying

Shows a real-world implementation of mobile, camera-driven pest detection and precise spraying—valuable as a technical and proof-of-concept benchmark.

[10] **Business Insider (2025):** *Spotta's IoT sensors and AI to reduce pesticide use*

Illustrates modern deployments of sensor-based, species-specific pest detection using IoT and AI—with benefits in early detection and reduced chemical usage.

Gantt Chart

